Topic 2.15 - Semi-log Plots

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________________ Class: ______________ Date: ______________

Why this topic matters

A semi-log plot makes exponential data linear by plotting the input against the logarithm of a positive output. Equal vertical distances represent equal ratios, not equal differences. The line's intercept determines the exponential coefficient, and its slope determines the growth or decay factor.

Learning targets

- Interpret semi-log plots with a logarithmically scaled output axis.
- Use linearity in x versus $\log y$ to identify and construct exponential models.
- Explain how slope, intercept, base, shifts, and limitations affect linearization.

Language and structure

Term	Meaning in this topic
semi-log plot	A graph with one logarithmic axis and one ordinary linear axis.
linearization	A transformation that turns a nonlinear relationship into a line.
log-output coordinate	The plotted value $z = \log y$ or $z = \ln y$ rather than y itself.
slope-factor link	If $\log_b y = mx + c$, then $y = b^c(b^m)^x$.
baseline shift	A limiting value L that must often be subtracted before taking a logarithm.

Launch: make a claim before calculating

Launch investigation

Reasoning first

On a base-10 logarithmic vertical axis, the labels 1, 10, 100, and 1000 are equally spaced. Why does this not mean the output changes by an equal amount between consecutive labels?

Concept 1: Why exponentials become lines

Core idea

If $y = ab^x$ with $a > 0$, then $\log y = \log a + x \log b$.

- On a plot of x versus $\log y$, the intercept is $\log a$.
- The slope is $\log b$ for the same log base used on the axis.
- A positive slope means growth; a negative slope means decay.

Worked example - annotate the reasoning

Problem. Linearize $y = 5(2)^x$ using common logarithms and identify the line's slope and intercept.
Model reasoning. Let $z = \log y$. Then $z = \log 5 + x \log 2$. The intercept is $\log 5 \approx 0.6990$ and the slope is $\log 2 \approx 0.3010$.

Check your understanding Independent

If $\ln y = 1.2 - 0.3x$, write y as an exponential function.

Concept 2: Read slope and intercept from a semi-log line

Core idea

If $z = \log_{10} y = mx + c$, then $a = 10^c$ and the one-unit factor is 10^m.

- A change of Δx multiplies y by $10^{m\Delta x}$.
- Vertical intervals labeled $1, 10, 100, 1000$ are equally spaced on a base-10 log axis.
- Use actual log coordinates rather than visual ordinary-axis distances.

Worked example - annotate the reasoning

Problem. A semi-log line satisfies $\log y = 0.7 + 0.15x$. Find the exponential model and interpret the one-unit factor.
Model reasoning. $y = 10^{0.7}(10^{0.15})^x \approx 5.012(1.413)^x$. Each one-unit increase in x multiplies y by about 1.413.

Check your understanding

Independent

On a base-10 semi-log plot, the line rises 0.6 log units when x increases by 4. Find the factor per unit.

Concept 3: Shifts, diagnostics, and limitations

Core idea

A shifted exponential $y = L + ab^x$ is linearized by plotting $\log(y - L)$, not usually by plotting $\log y$.

- All transformed outputs must be positive.
- Curvature on a semi-log plot suggests that a single unshifted exponential is not adequate.
- Linearity over a short interval does not guarantee reliable extrapolation.

Worked example - annotate the reasoning

Problem. Explain how to linearize $T(t) = 20 + 70(0.8)^t$.
Model reasoning. Subtract the baseline: $T - 20 = 70(0.8)^t$. Then $\ln(T - 20) = \ln 70 + t \ln 0.8$, which is linear in t.

Check your understanding

Independent

A semi-log plot bends downward rather than forming a line. Give two possible explanations.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

If $\log y = 0.4x + 1$, write y in exponential form.

Exit ticket 2

No calculator

What is the semi-log slope for $y = 12(0.7)^x$ using natural logarithms?

Exit ticket 3

No calculator

Why must $y > 0$ before plotting $\log y$?